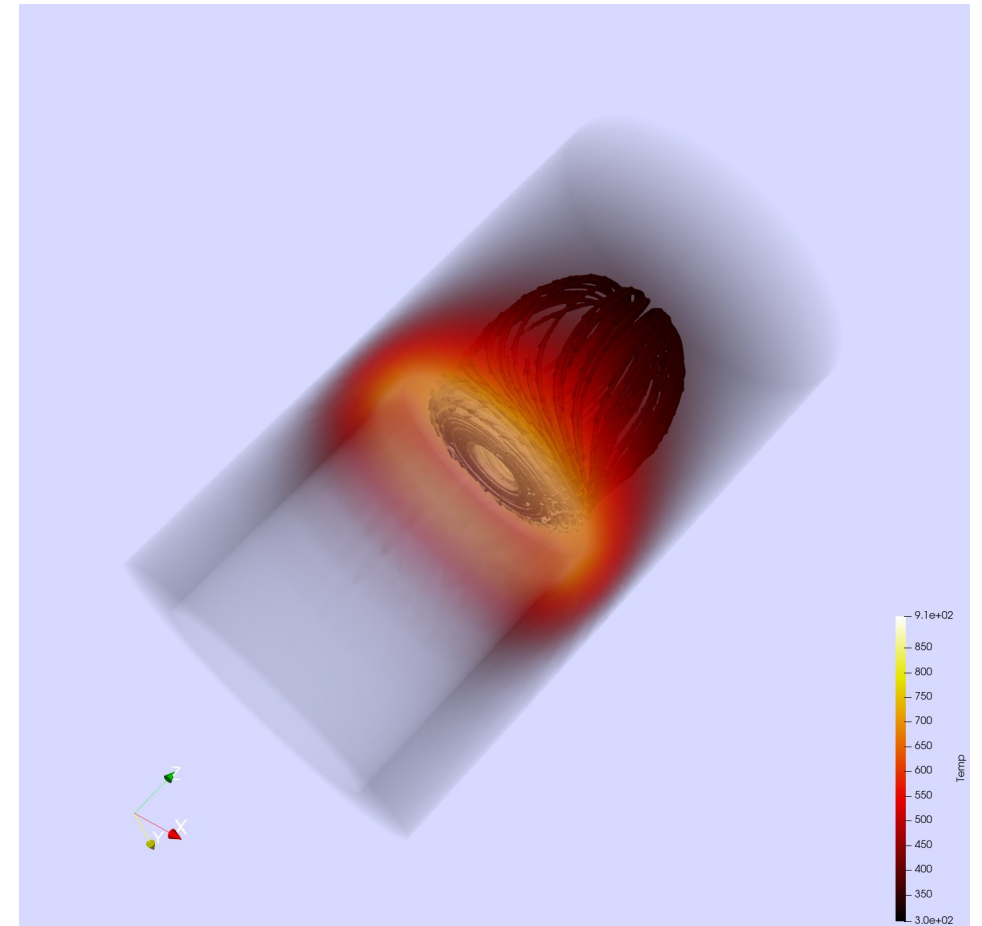

Direct Volume Rendering

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2025



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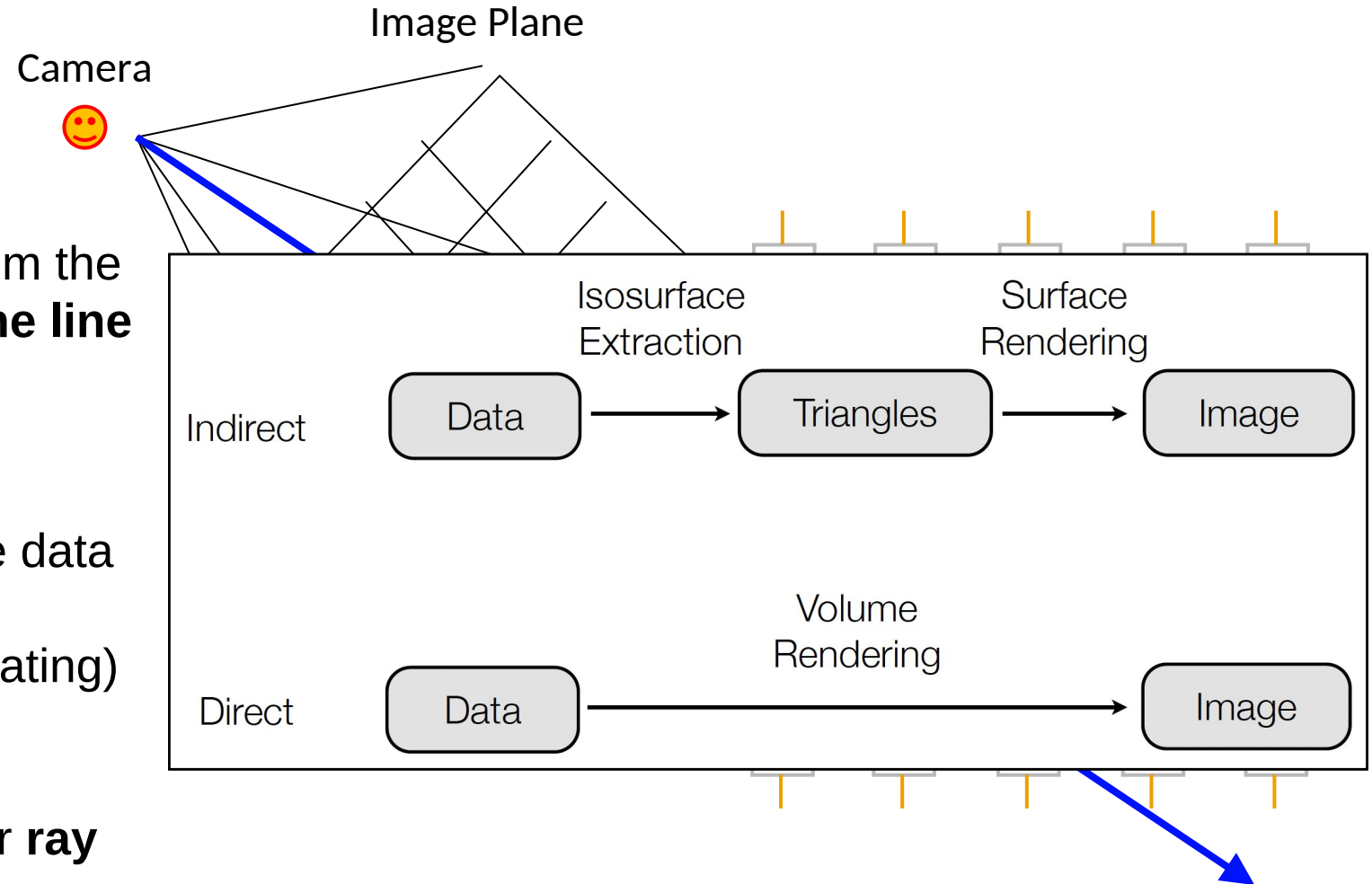
Direct Volume Rendering: Concept

Generate a 2D image **directly** from the 3D data by **compositing** along the **line of sight**:

directly – without generating an intermediate representation of the data

compositing - combining (integrating) data along the viewing direction

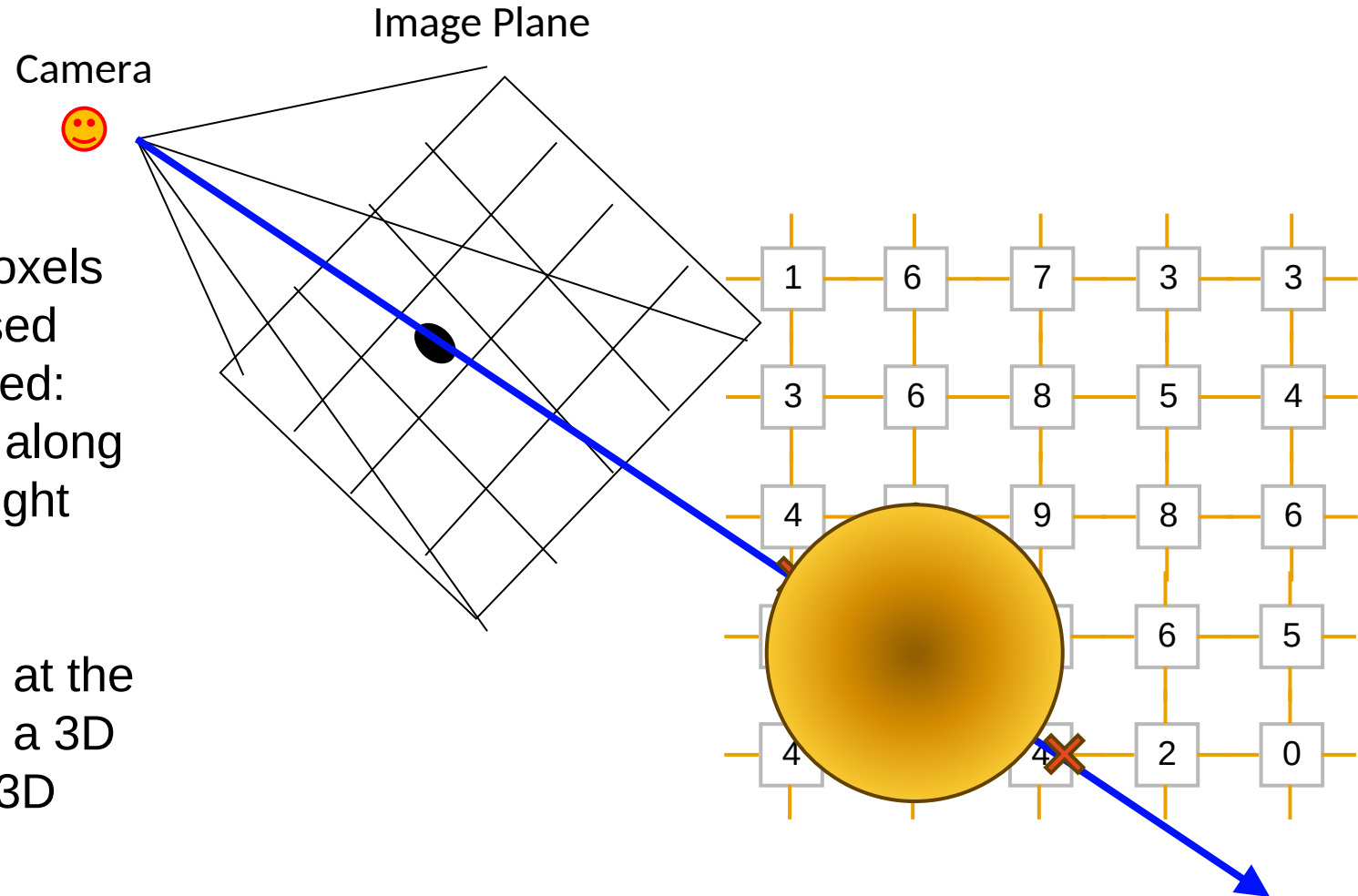
line of sight - viewing direction or **ray**





Direct Volume Rendering: Steps

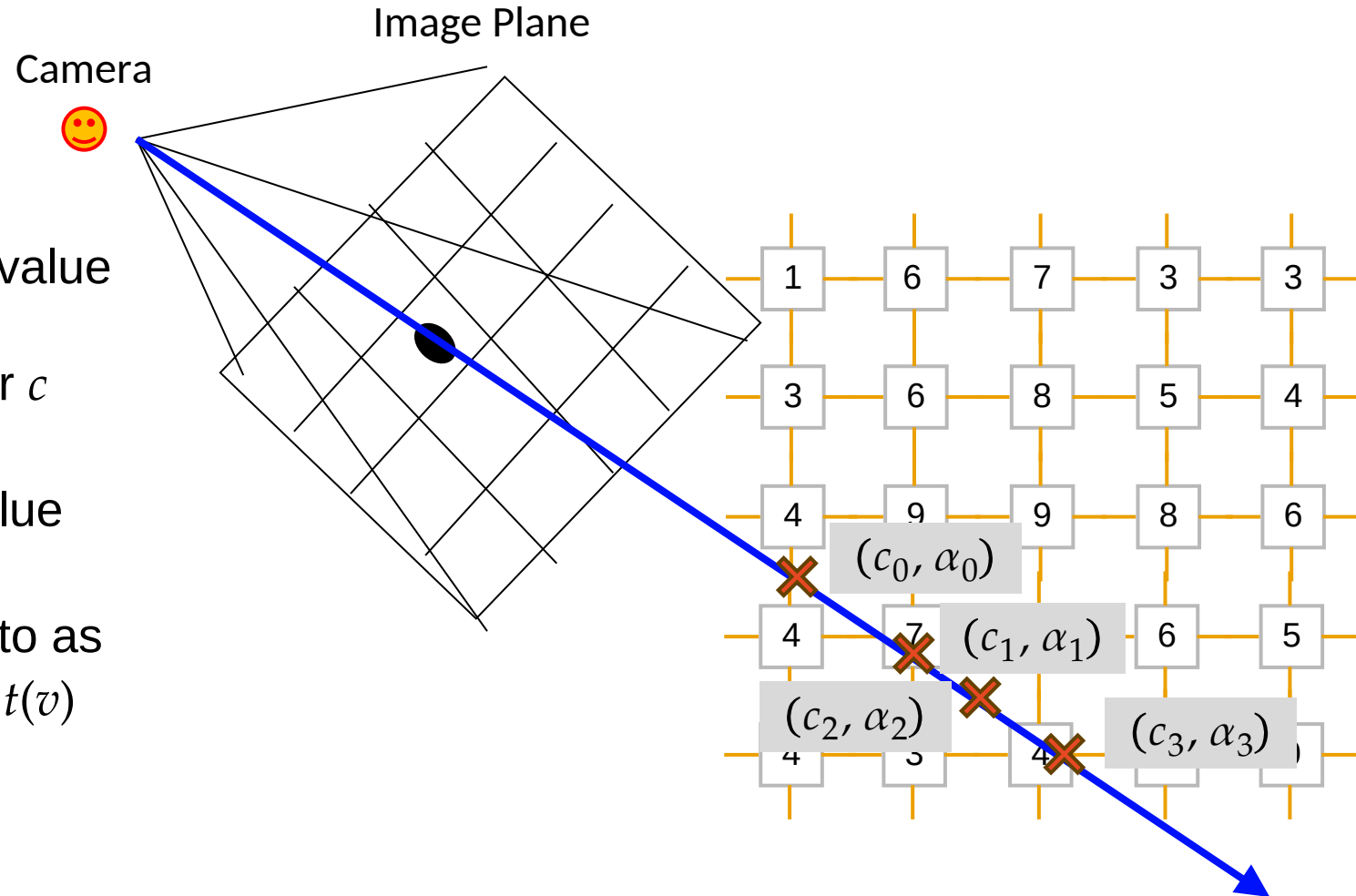
1. **sampling** : selecting which voxels in the discrete data set are used and how their data is combined:
 - a. **ray marching** : integrate along the ray in steps (which might not be uniform)
 - b. **interpolation kernel** : interpolate the data value at the point of interest based on a 3D neighborhood (e.g., box, 3D Gaussian)





Direct Volume Rendering: Steps

2. **classification** : sample data value is used to evaluate optical properties, such as RGB color c and opacity α ;
the mapping from the data value $v = \mathcal{D}(x,y,z)$ to the optical properties c and α is referred to as the **transfer function**: $(c, \alpha) = t(v)$



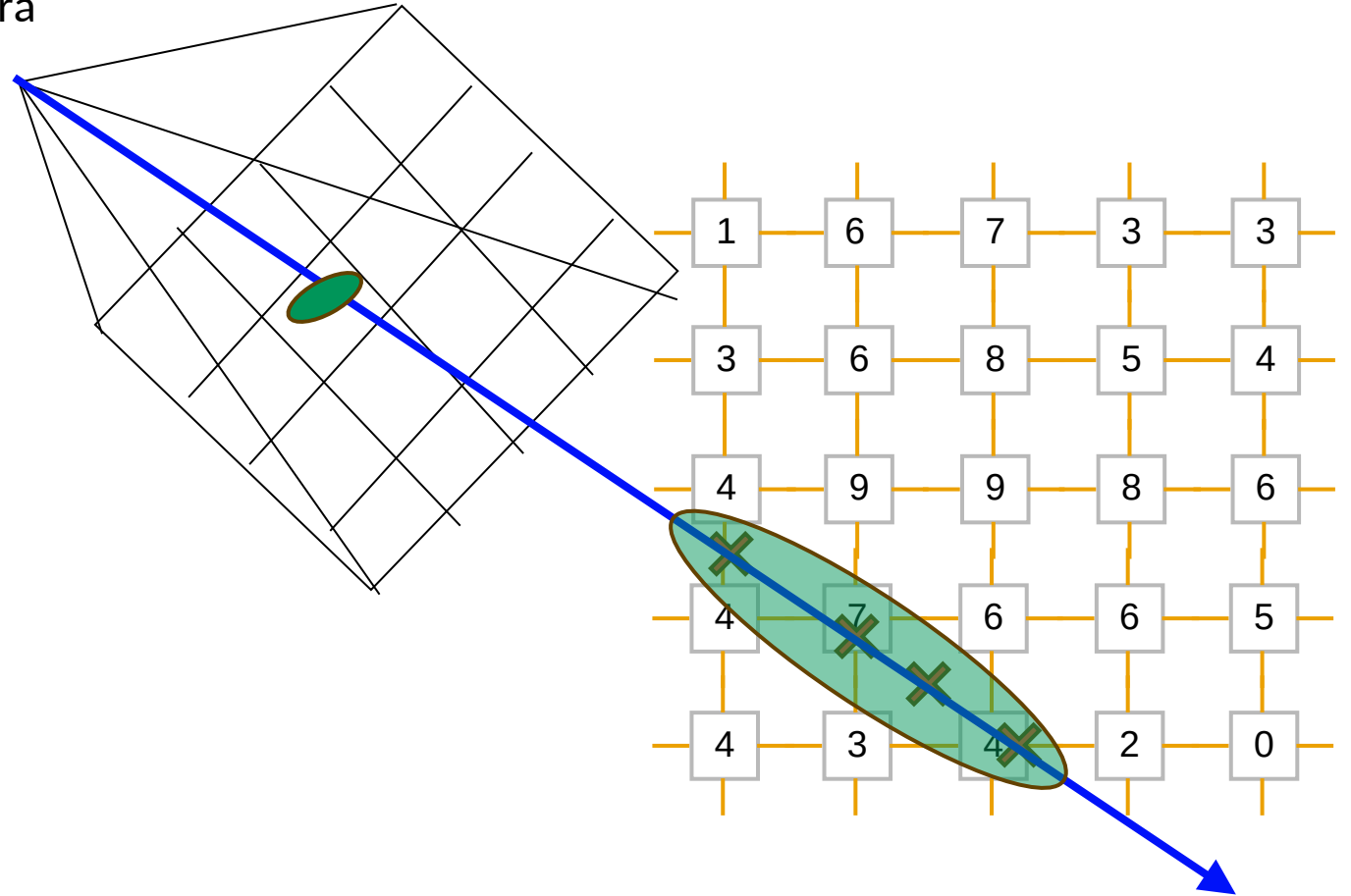


Direct Volume Rendering: Steps

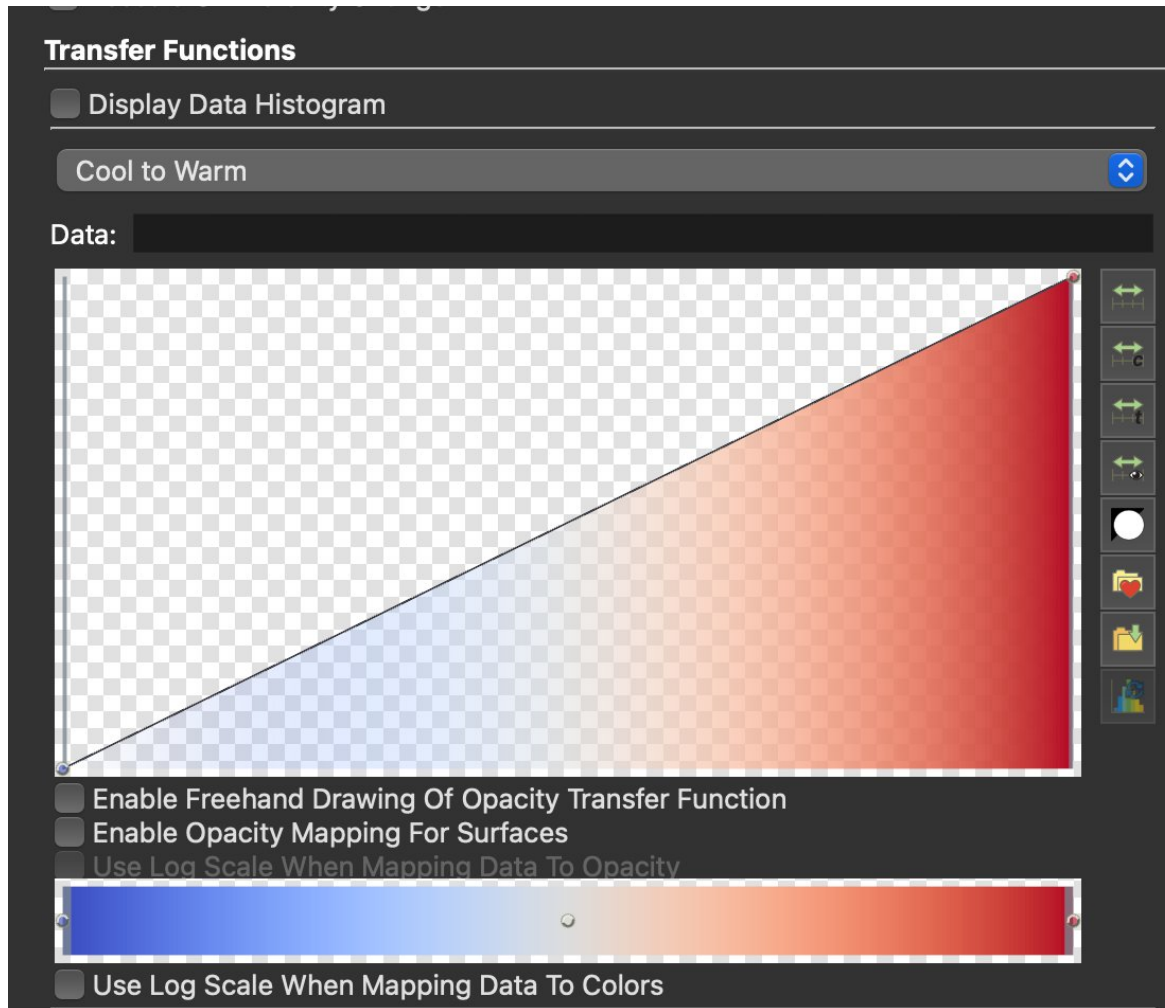
3. **compositing** : evaluating the color for the line of sight by blending the optical properties:

$$I = f((c_0, \alpha_0), \dots, (c_{n-1}, \alpha_{n-1}))$$

Camera

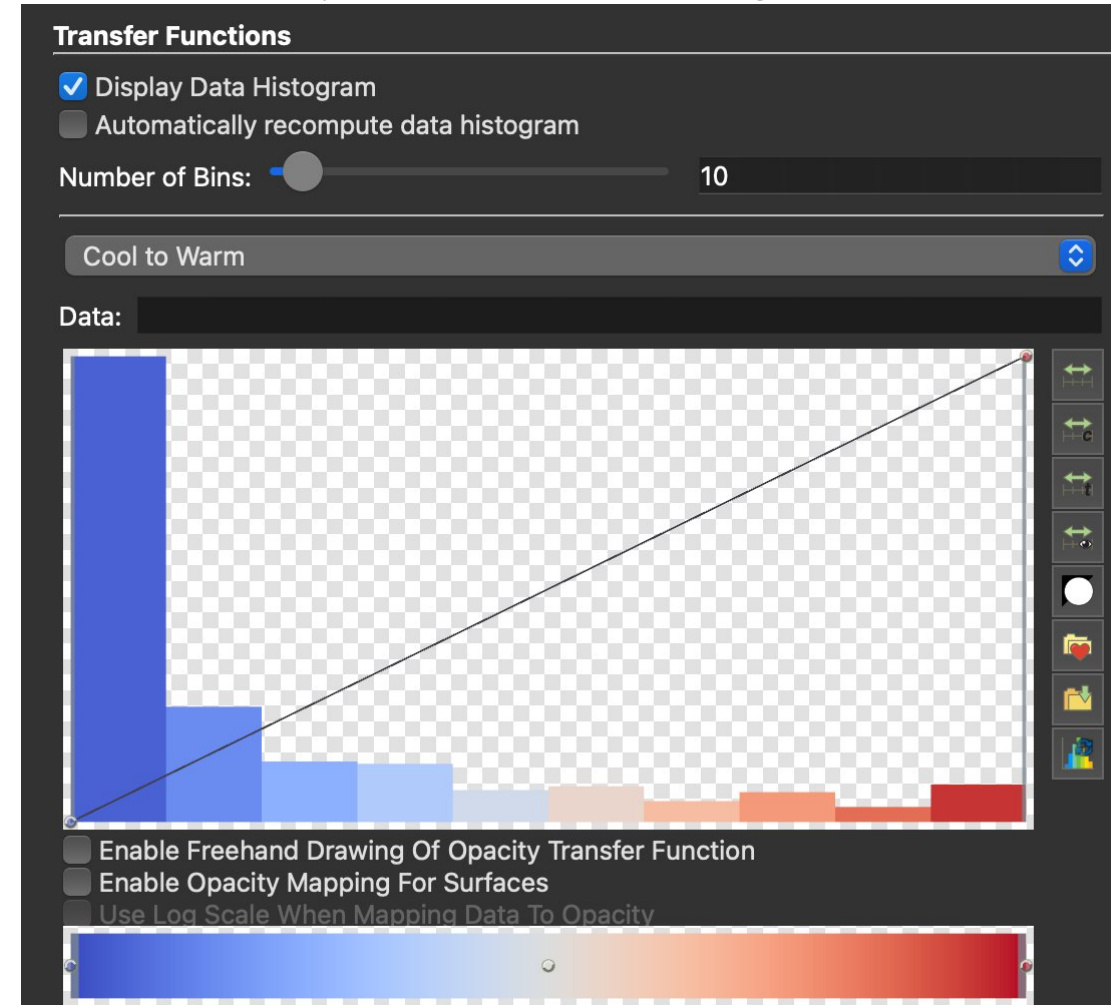


Transfer Function



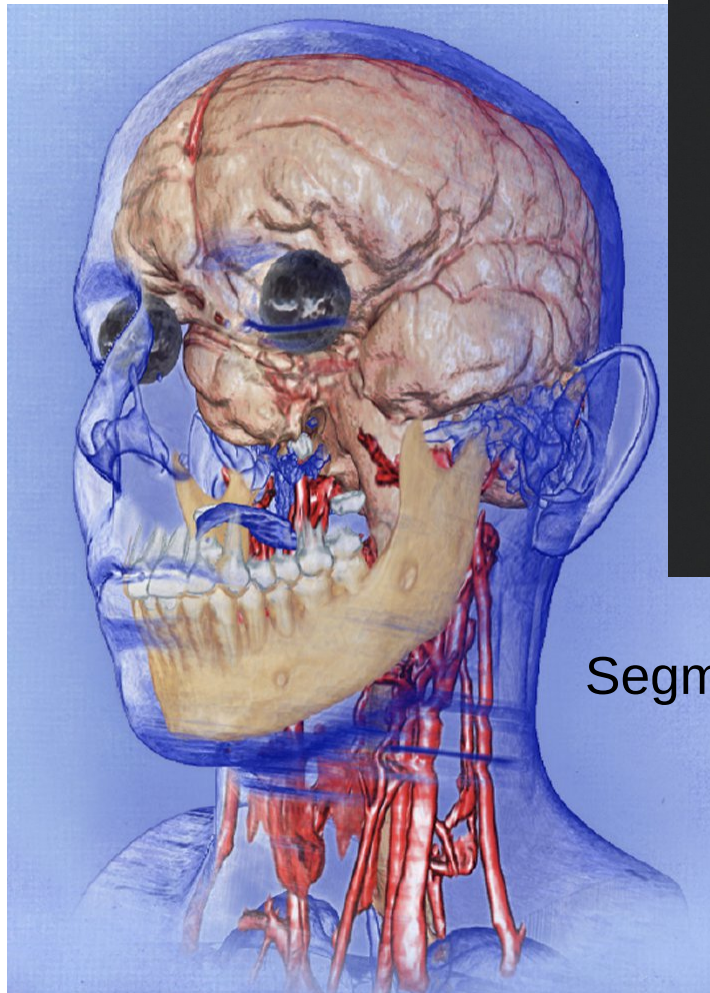
- Maps a data value to RGBA

$$f:\mathbb{R} \longrightarrow \mathbb{R}^4, v \longrightarrow (r,g,b,\alpha)$$

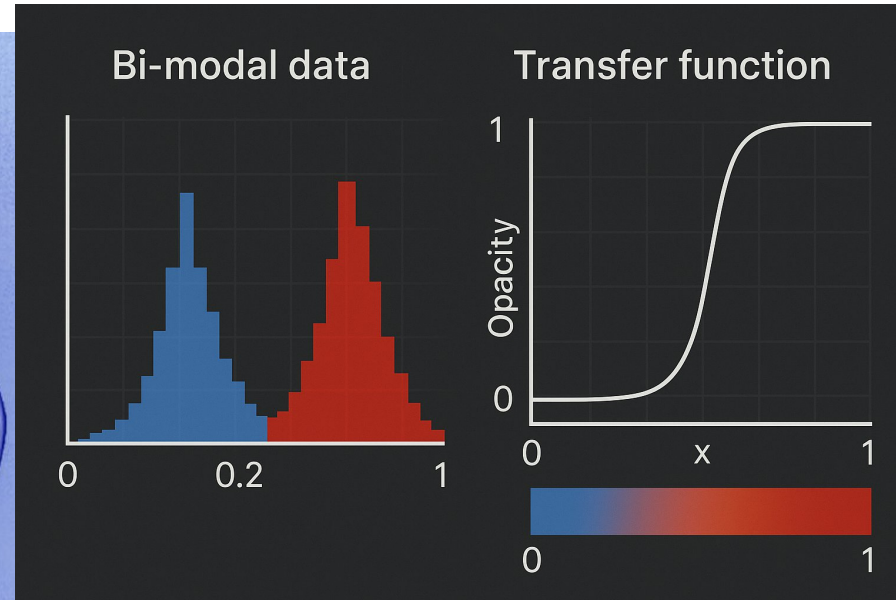




Transfer Function



Segmentation



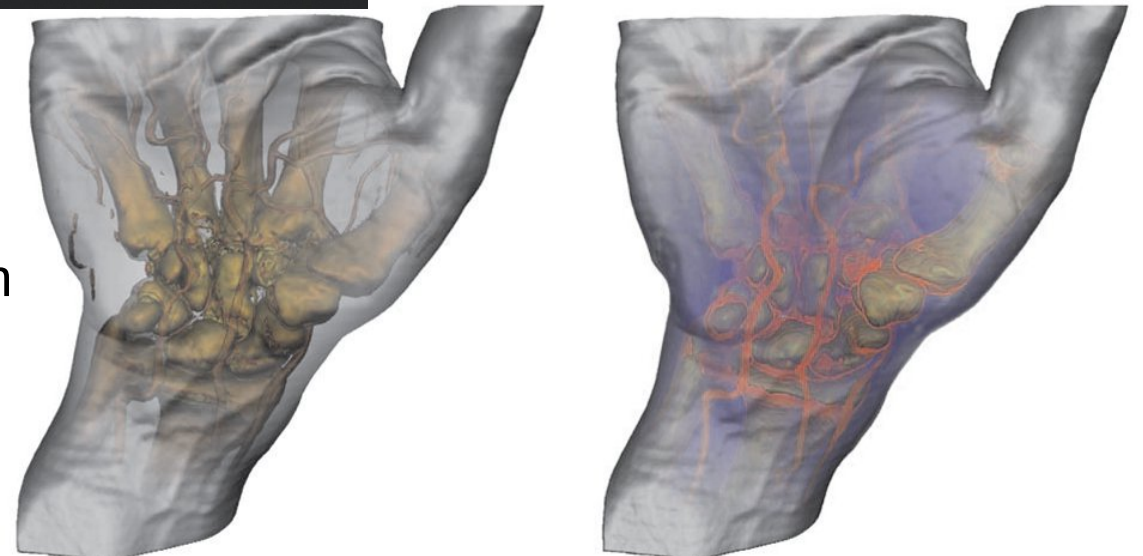
“State of the Art in Transfer Functions for Direct Volume Rendering”

Patric Ljung, Jens Krueger, Eduard Groeller, Markus Hadwiger, Charles D. Hansen and Anders Ynnerman

Computer Graphics Forum, 35(3), pp. 669-691, 2016

<https://doi.org/10.1111/cgf.12934>

Exploration

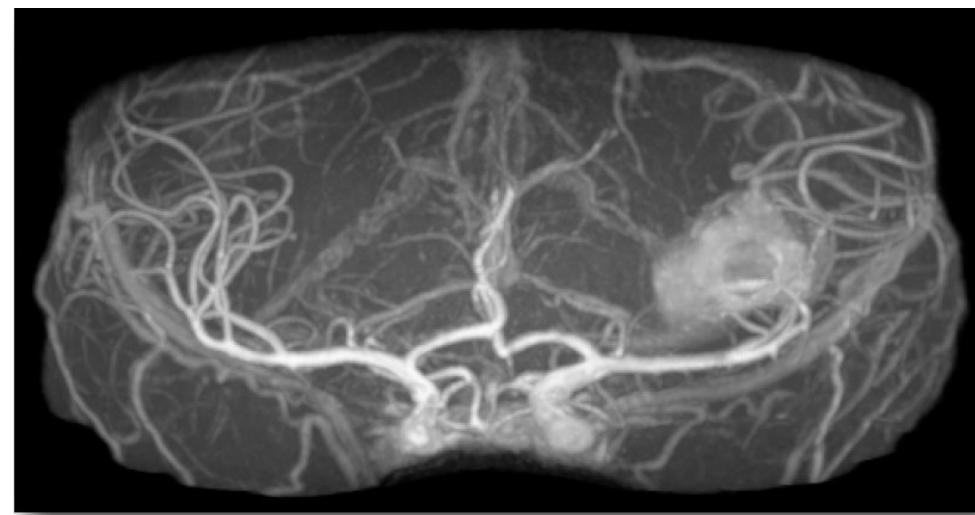
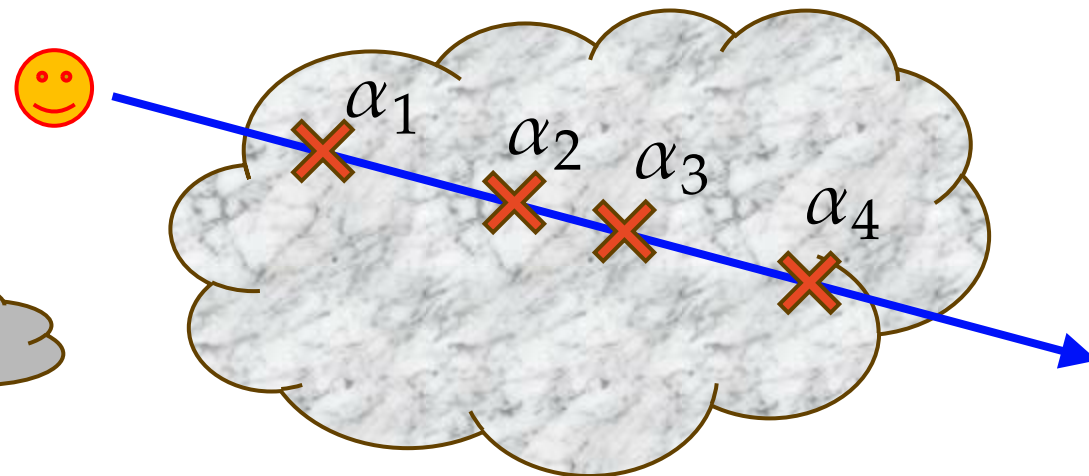


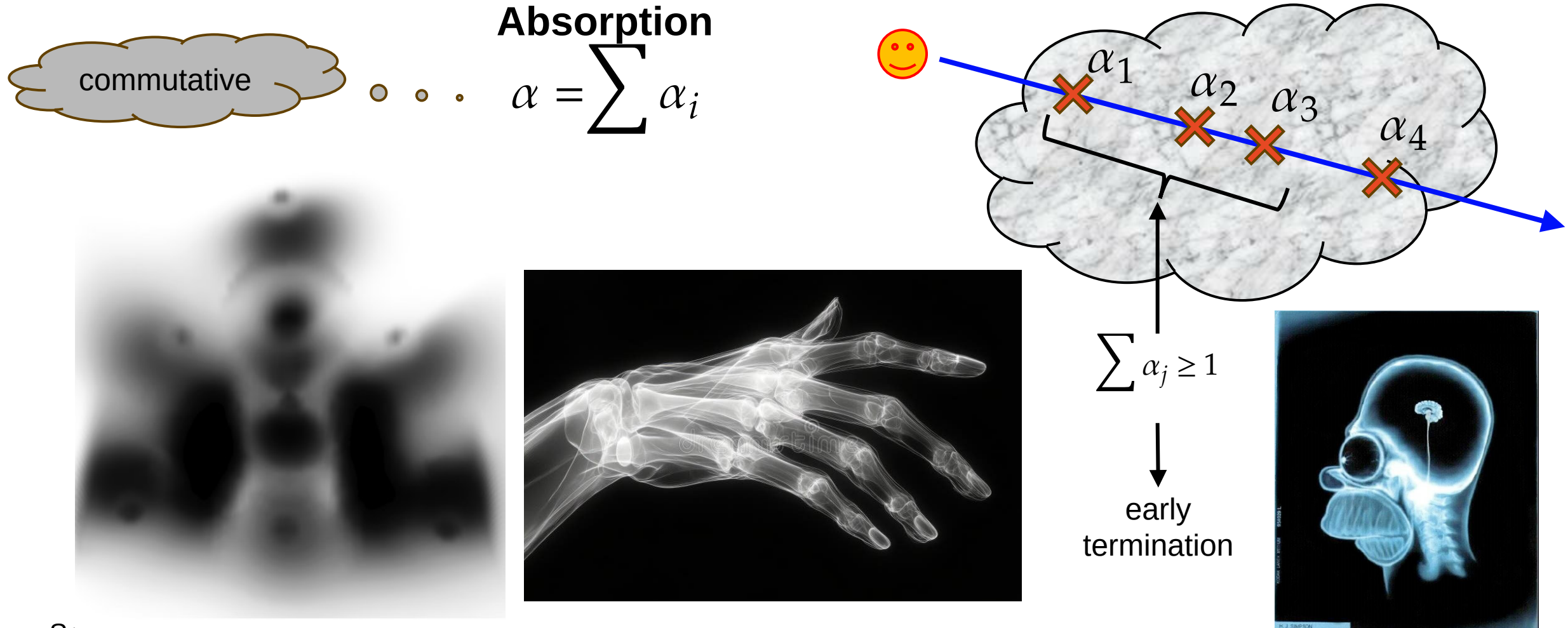


Maximum Intensity Projection (MIP)

$$\alpha = \max (\alpha_1, \dots, \alpha_{n-1})$$

commutative





Absorption + Emission

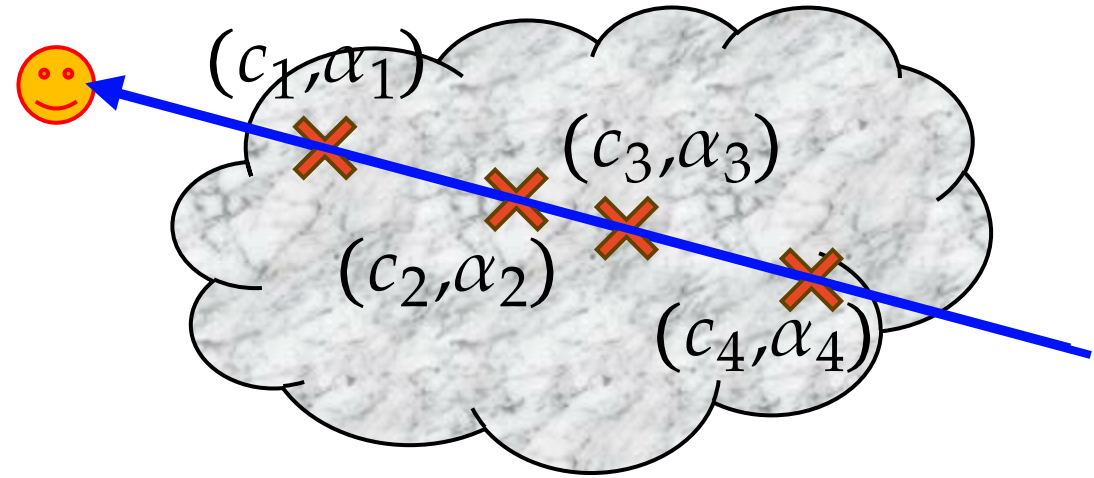
- Besides opacity α_i each point also emits diffuse light c_i
- This optical models is not commutative: order matters!

Back to Front

Let C_i be the colour at point i due to all points $j \geq i$

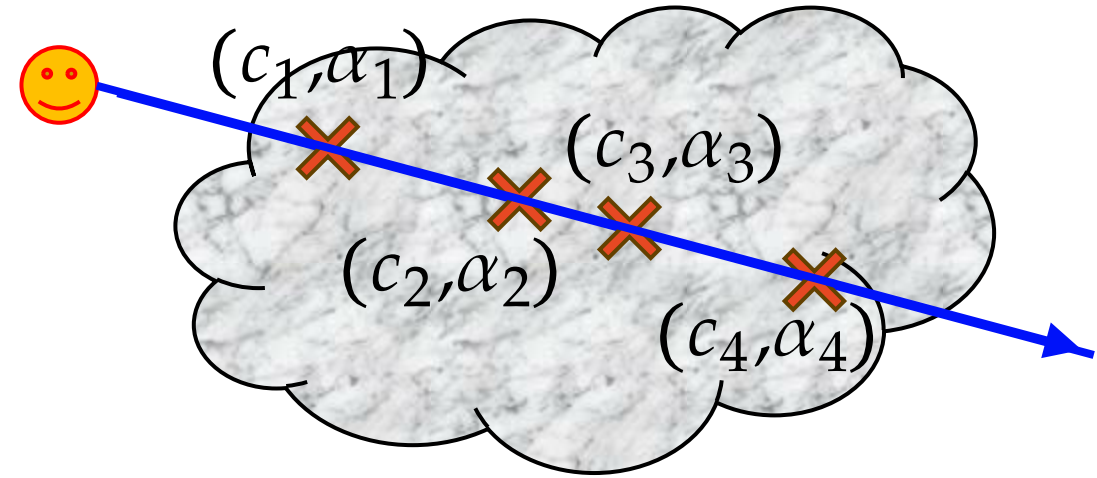
$$C_4 = c_4\alpha_4$$

$$C_i = c_i\alpha_i + C_{i+1}(1-\alpha_i)$$



Absorption + Emission

- Besides opacity α_i each point also emits diffuse light c_i
- This optical models is not commutative: order matters!



Front to Back

Let A_i be the accumulated opacity due to all points $j \leq i$

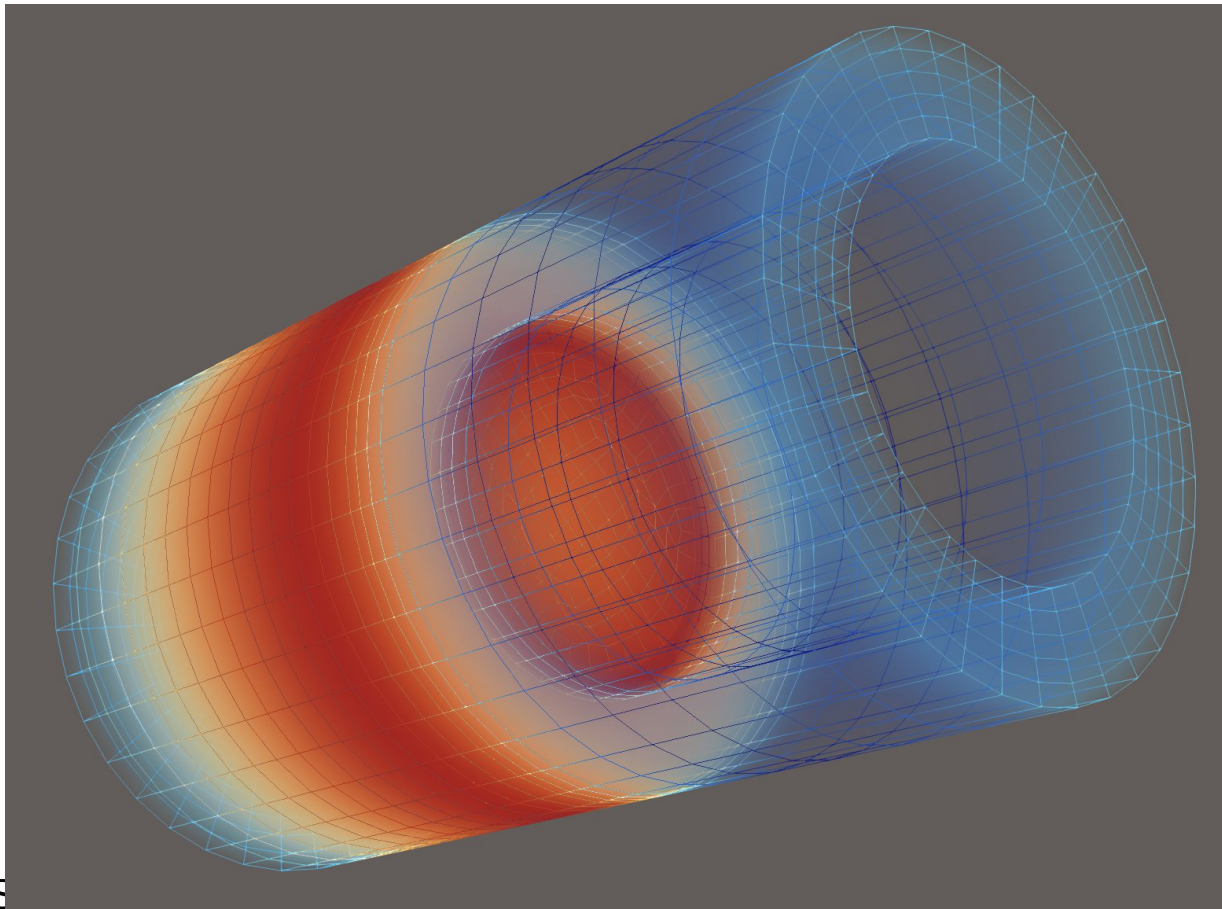
$$C_0 = A_0 = 0$$

$$C_i = C_{i-1} + (1 - A_{i-1}) \alpha_i c_i$$

$$A_i = A_{i-1} + (1 - A_{i-1}) \alpha_i$$



Absorption + Emission



Visualization

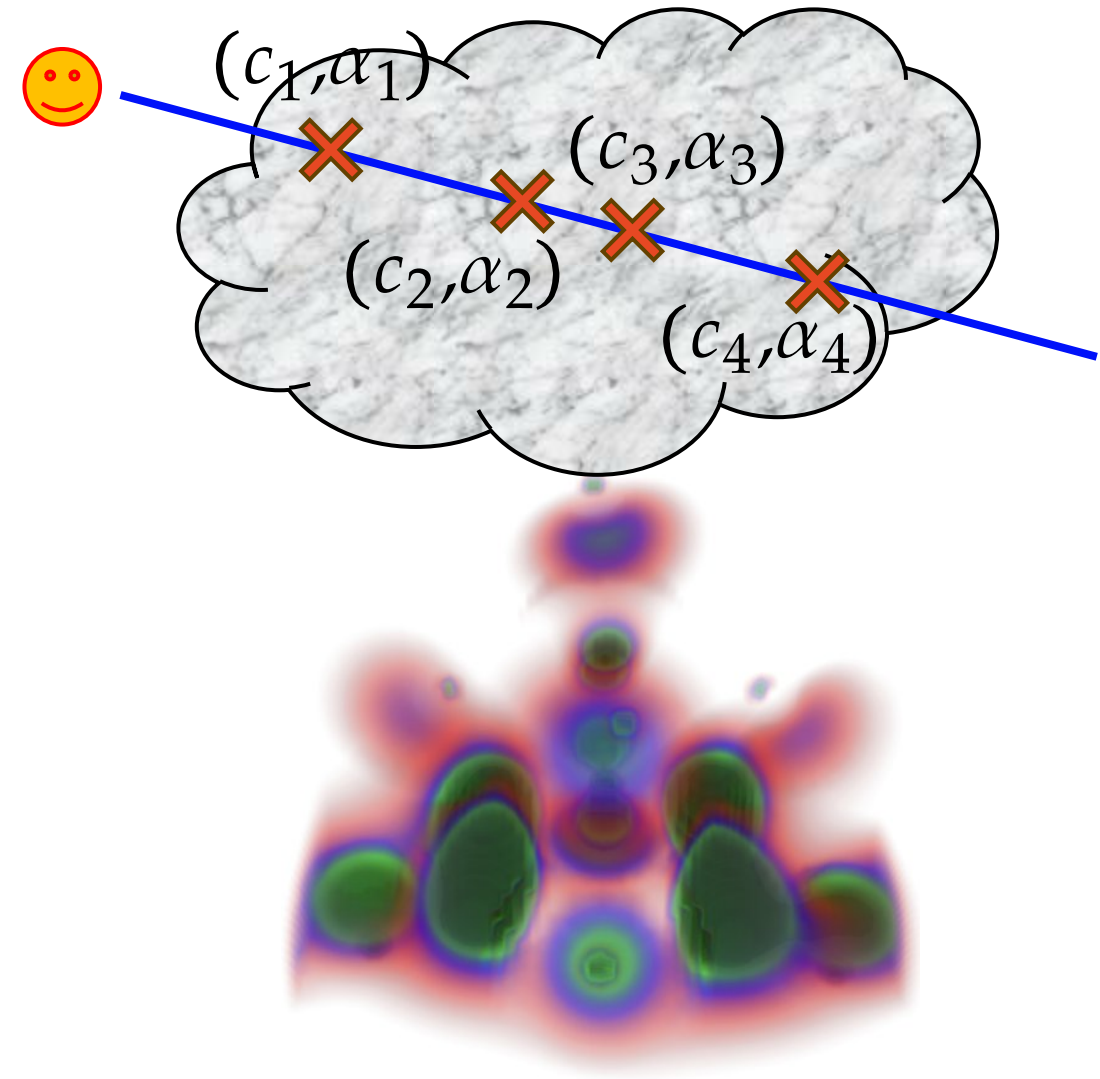
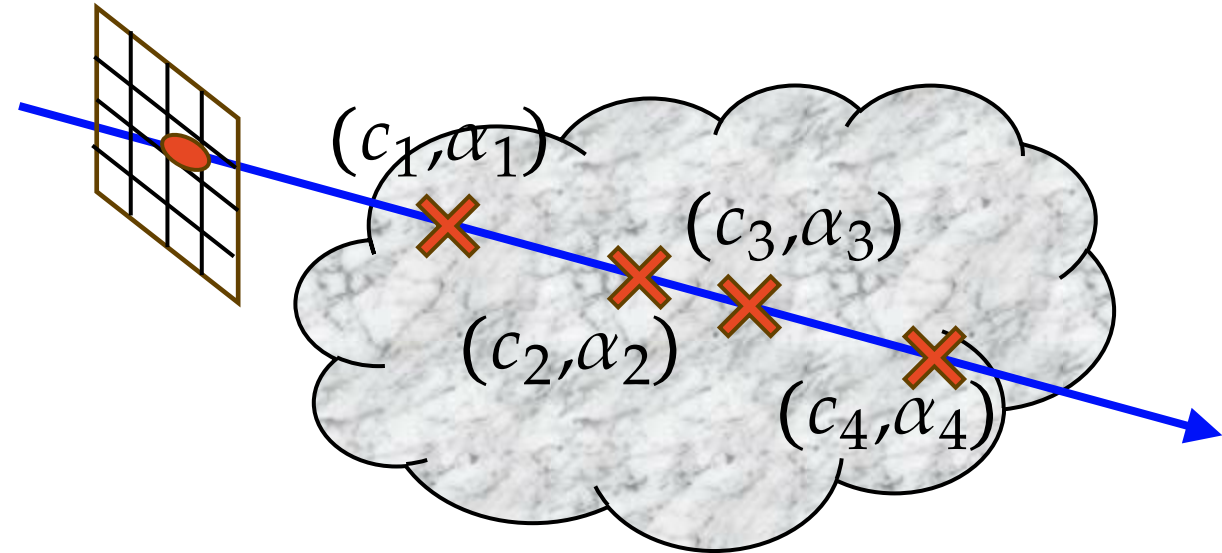


Image order, front to back algorithm



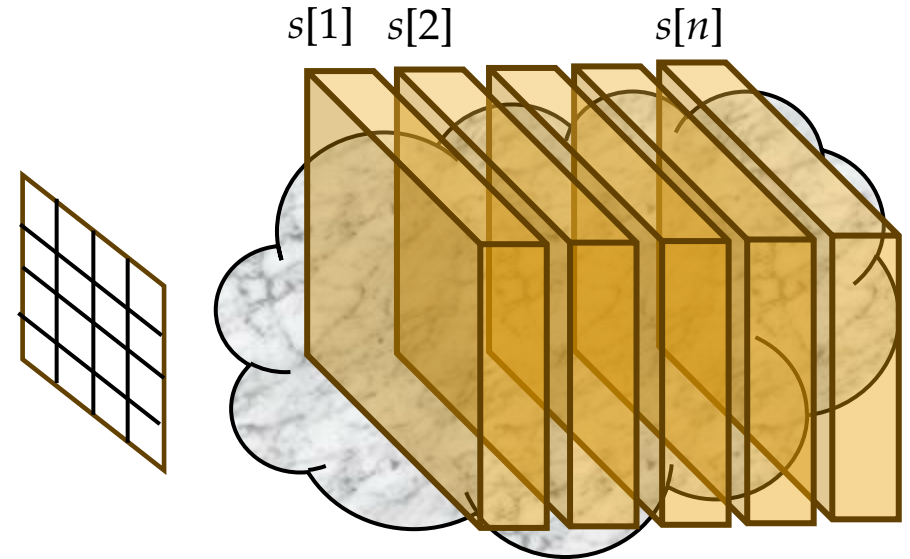
1. for each point p in the image plane
 1. cast a ray from the camera through p
 2. $C_0 = A_0 = 0$
 - // ray marching
 3. for each step $i = 1 \dots n$ along the ray
 1. interpolate the data value v_i from the neighborhood
 2. $(c_i, \alpha_i) = t(v_i)$ // transfer function
 3. $C_i = C_{i-1} + (1 - A_{i-1}) \alpha_i c_i$
 4. $A_i = A_{i-1} + (1 - A_{i-1}) \alpha_i$
 5. if $A_i \geq 1$ break // early termination
 4. $I(p) = C_i$

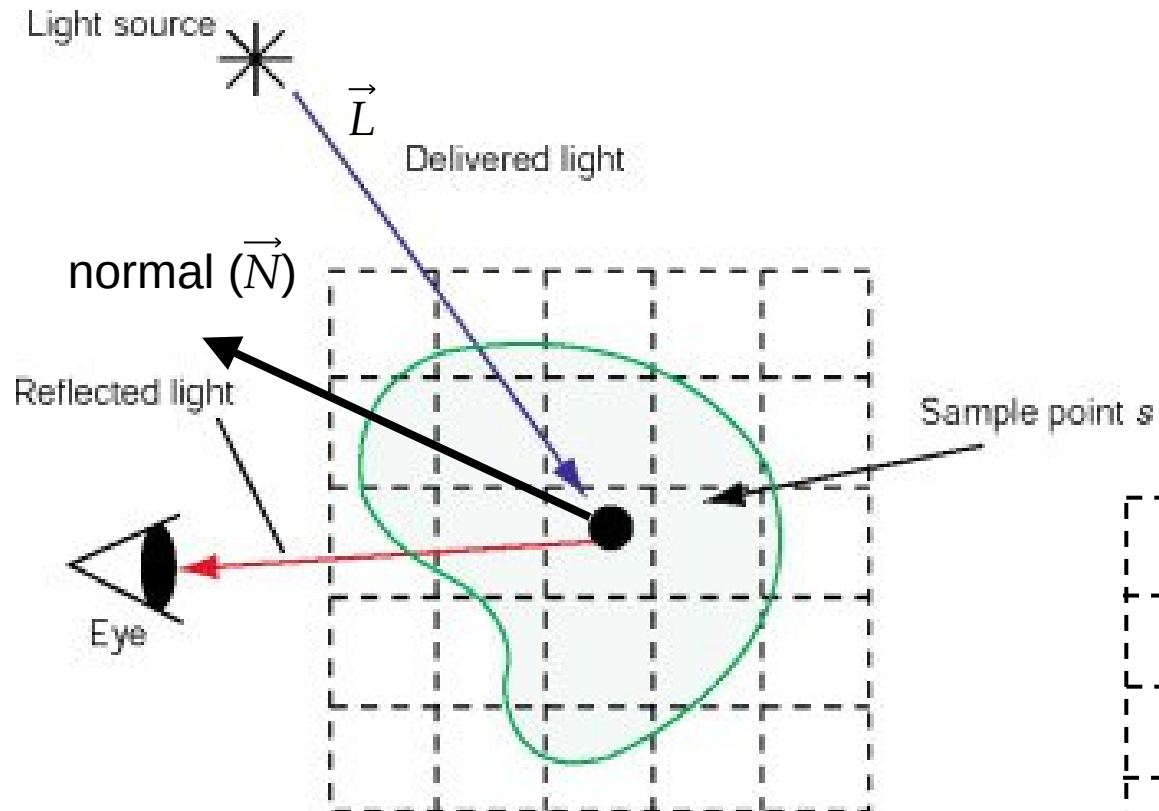




Object order, back to front algorithm

1. for all points p in the image plane $I[p] = 0$
2. for all data slices $i = n \dots 1$
 1. $s = \text{data slice } s[i]$
 2. project all voxels in s onto the image plane
 3. for each point p in the image plane
 1. $(c_i, \alpha_i) = t(v_i)$ // transfer function
 2. $I[p] = c_i \alpha_i + I[p] (1 - \alpha_i)$



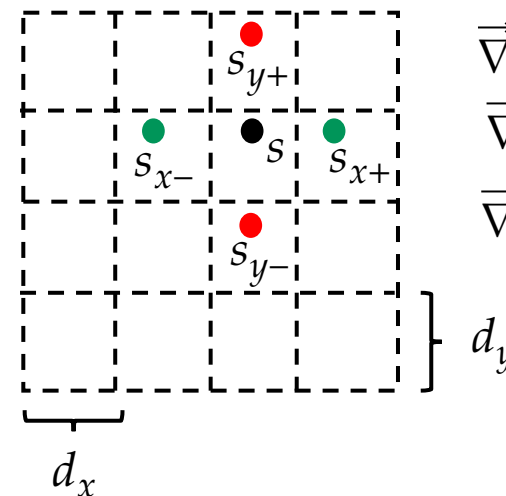


Lambert attenuation factor:

$$L(s) = \max(\vec{N} \cdot \vec{L}, 0)$$

Use the gradient as the normal:

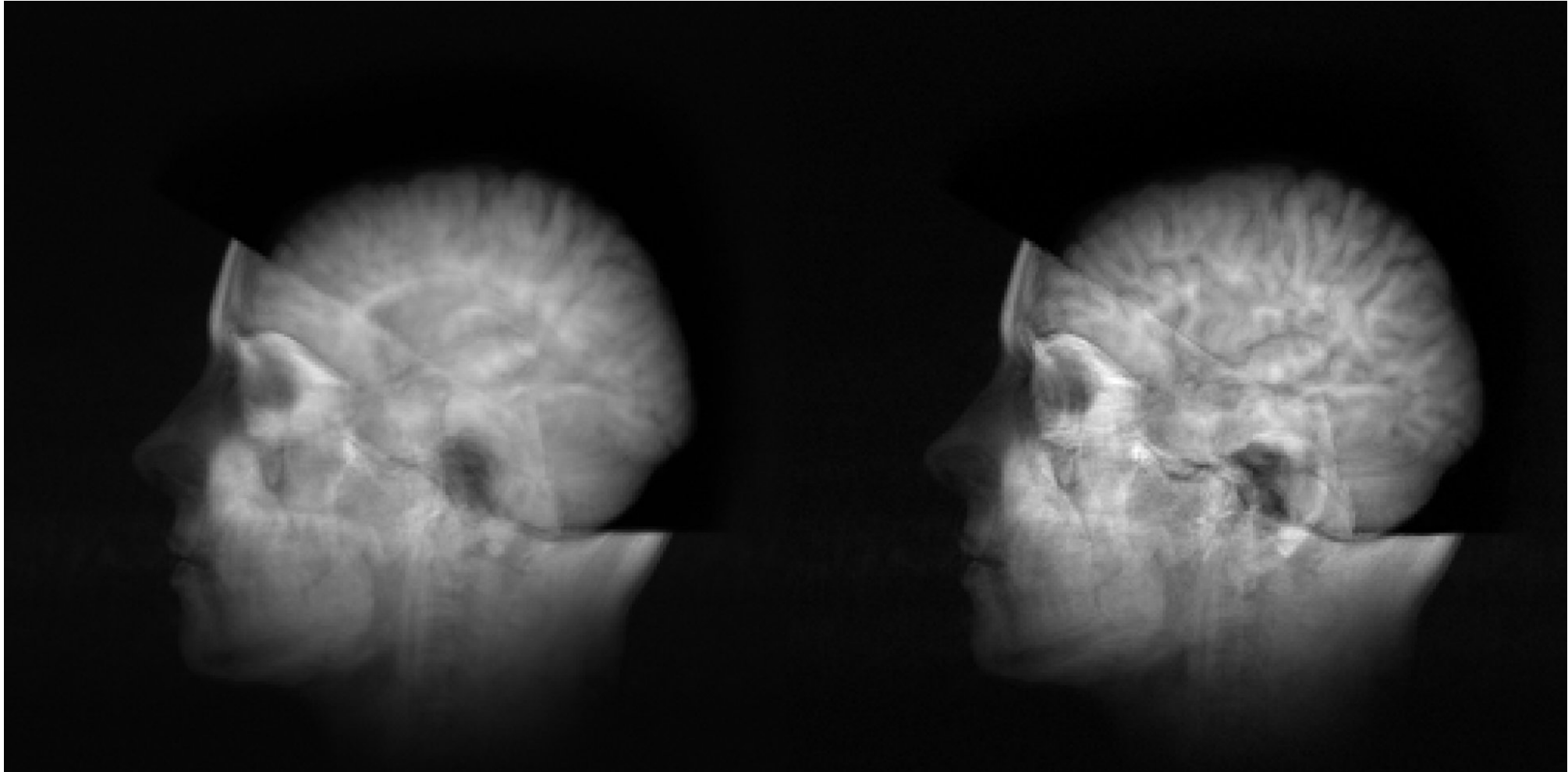
$$\vec{N} = \vec{\nabla} f(s) / \|\vec{\nabla} f(s)\|$$



$$\vec{\nabla} f_x(s) = (f(s_{x+}) - f(s_{x-})) / (2d_x)$$

$$\vec{\nabla} f_y(s) = (f(s_{y+}) - f(s_{y-})) / (2d_y)$$

$$\vec{\nabla} f_z(s) = (f(s_{z+}) - f(s_{z-})) / (2d_z)$$



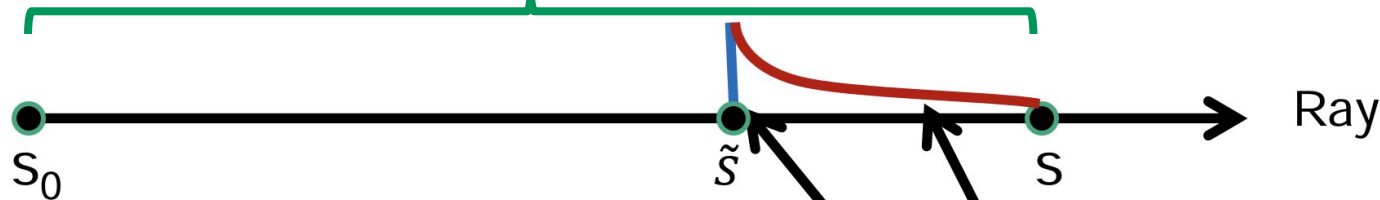


Physically based volume rendering

Optical depth τ
Absorption κ

$$\tau(s_1, s_2) = \int_{s_1}^{s_2} \kappa(s) ds$$

integrates all points along the ray



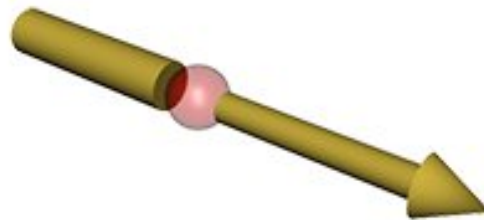
$$I(s) = I(s_0)e^{-\tau(s_0,s)} + \int_{s_0}^s q(\tilde{s})e^{-\tau(\tilde{s},s)} d\tilde{s}$$

Emission at $\tilde{s}$

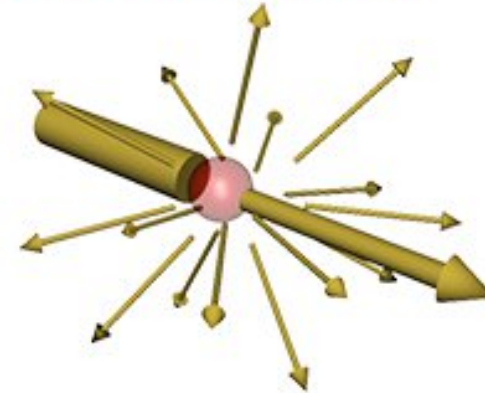
Attenuation along $\tilde{s} - s$



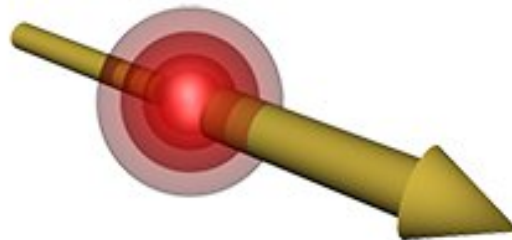
Absorption



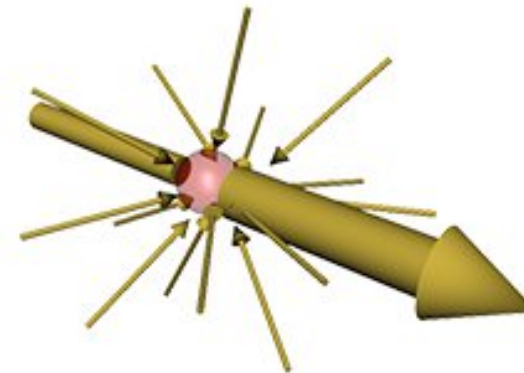
Out-scattering



Emission



In-scattering



(c) www.scratchapixel.com

